Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	14 October 2026
Team ID	
Project Name	Automated Network Request Management System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable includes the architectural diagram below and the information as per Table 1 & Table 2:

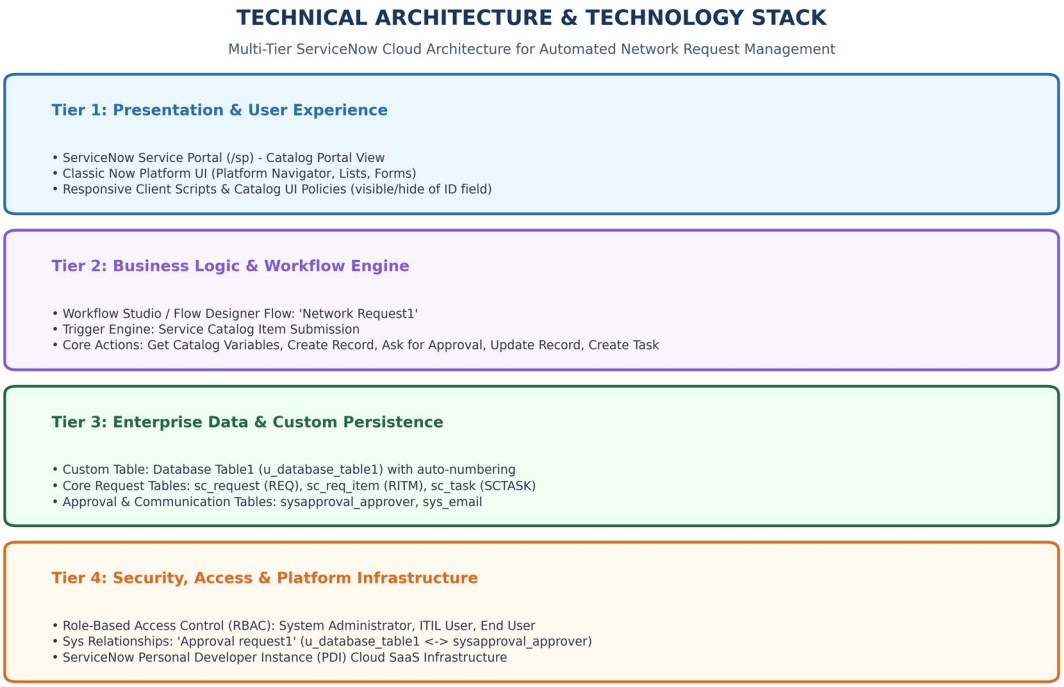


Figure 2.4: Multi-Tier ServiceNow Technical Architecture

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Self-service web intake portal for network requests	ServiceNow Service Portal (/sp) & Classic UI

2	Service Catalog	Catalog item capturing contact, connection, amount, address	Service Catalog (sc_cat_item: 'Network Request1')
3	Client Validation	Conditional visibility and mandatory checking on Existing ID	Catalog UI Policy ('visible/hide of ID field')
4	Workflow Engine	Low-code orchestration coordinating trigger, data, approvals	Workflow Studio / Flow Designer ('Network Request1')
5	Data Persistence	Custom table persisting request details independently of cart	Database Table1 (u_database_table1)
6	Approval Engine	Multi-tier approval routing for managers and administrators	Flow Action: Ask for Approval & sysapproval_approver
7	System Relationship	Custom query script linking approval records to custom rows	System Relationship ('Approval request1' on sys_relationship)
8	Notification Engine	Automated HTML email generation on creation and approval	Flow Action: Send Email & ServiceNow sys_email

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Cloud Infrastructure	Multi-tenant cloud platform hosting instance dev251577	ServiceNow Enterprise Cloud PaaS/SaaS
2	Security Implementation	Role-Based Access Control protecting admin functions	RBAC: System Administrator, ITIL User, End User
3	Scalable Architecture	Decoupled multi-tier service management architecture	ServiceNow Multi-Tier Service Architecture
4	Relational Data Model	Referential integrity, data pills, and auto-numbering	MariaDB-backed ServiceNow DB (sys_db_object)
5	Audit & Compliance	Automated activity stream, timestamps, and email logs	Built-in Sys Audit, Activity Stream, sys_email